

Names: _____
Section Number: _____

1. What is one thing you all have in common?
2. What are your strengths as a member of this team? Answer separately for each member.
3. How will you set up your group to ensure everybody contributes effectively? (For example, this could involve establishing group roles, or deciding on communication norms, or something else.)
4. How will you check up on your group dynamics throughout the semester?
5. How will you internally resolve problems that arise between group members?

Goal. Review u-substitution and use the integral to find the area of various regions. In each problem, be sure to draw a graph of the functions.

1. Evaluate $\int \cot(2x) \, dx$. (Hint: Re-write cotangent in terms of sine and cosine.)

2. Evaluate $\int_0^4 \frac{x}{\sqrt{x^2 + 1}} \, dx$.

3. Find the area enclosed between $y = \sin(x)$, $y = \cos(x)$, $x = 0$, and $x = \pi$.

4. Find the area of the region bounded by $y = e^x$, $y = 14 - e^x$, and the y -axis.

5. Find the area of the region enclosed between $x = y^2 - 5$ and $x = 3 - y^2$.